Practice workbook · Identify skin structures and state what each does

Naming skin layers and parts from a diagram or context and giving their roles: protection, sensing, melanin, fat insulation, barrier to infection.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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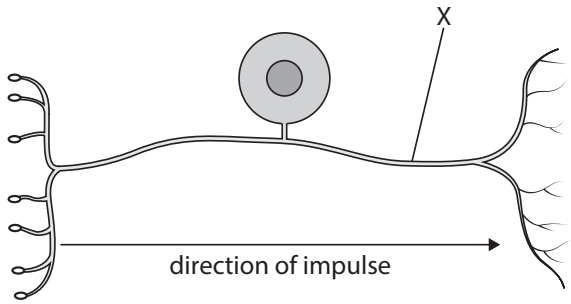
Type ID: HBI-SKIN-STRUCTURE-FUNCTION-RECALL

1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

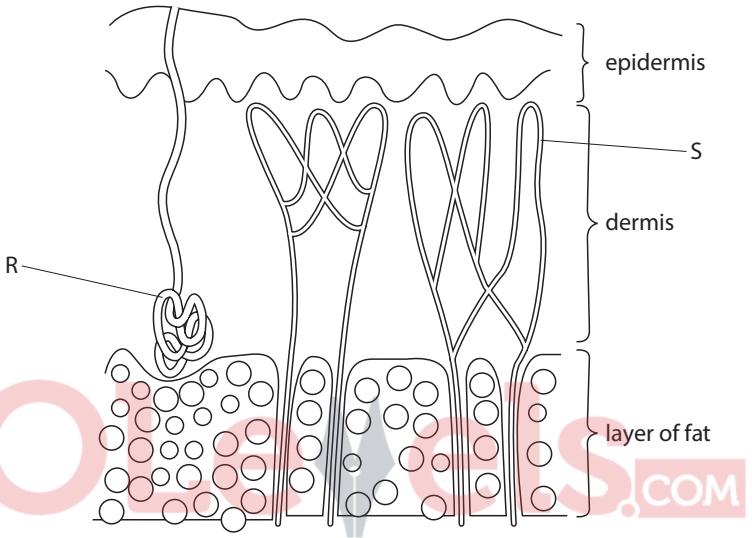
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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

.....

.....

.....

.....

1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

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(c) A student investigates the effect of temperature on receptors in the skin.

This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

(1)

(ii) Describe how to reduce the risk of this hazard.

(2)

(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

(1)

(Total for Question 1 = 12 marks)

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1. Question 1 · part b.iii · 2 marks · Working and answer

Working and answer

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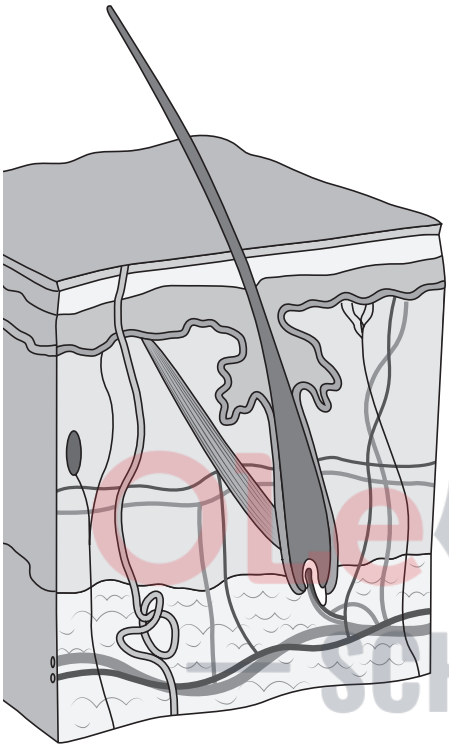
2. Question 5 · part a · 3 marks

January 2020 Paper 2

Further practice · Attempt part a only.

5 Burns can be first degree, second degree or third degree depending on how many layers of the skin are damaged.

The diagram shows the structure of the skin and the layers damaged by each type of burn.



layer 1	first degree burn
layers 1 and 2	second degree burn
layers 1, 2 and 3	third degree burn

(a) Name the three layers of skin that are damaged by a third degree burn.

(3)

layer 1 .....

layer 2 .....

layer 3 .....

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2. Question 5 · part a · 3 marks

January 2020 Paper 2

Further practice · Attempt part a only.

(b) Which two skin structures are damaged by a second degree burn?

(1)

- ☒ **A** blood vessels and sebaceous glands
- ☒ **B** fat tissue and blood vessels
- ☒ **C** hair follicles and sweat glands
- ☒ **D** sebaceous glands and fat tissue

(c) Explain why a person with a second or third degree burn has a high risk of infection.

(2)

(d) Blood clotting helps to prevent infection.

Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

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2. Question 5 · part a · 3 marks · Working and answer

Working and answer

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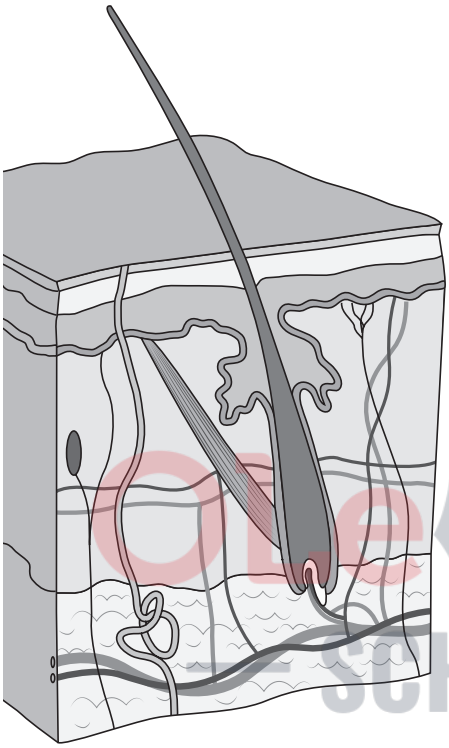
3. Question 5 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

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layer 2 .....

layer 3 .....

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3. Question 5 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

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Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

3. Question 5 · part b · 1 mark · Working and answer

Working and answer

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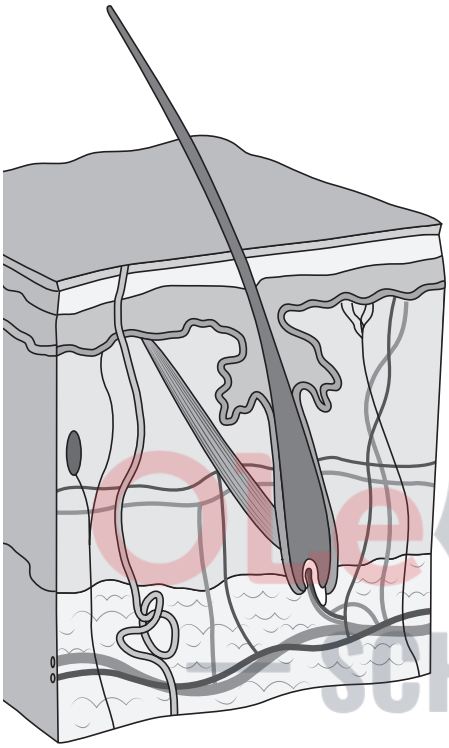
4. Question 5 · part c · 2 marks

January 2020 Paper 2

Further practice · Attempt part c only.

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The diagram shows the structure of the skin and the layers damaged by each type of burn.



layer 1	first degree burn
layers 1 and 2	second degree burn
layers 1, 2 and 3	third degree burn

(a) Name the three layers of skin that are damaged by a third degree burn.

(3)

layer 1 .....

layer 2 .....

layer 3 .....

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4. Question 5 · part c · 2 marks

January 2020 Paper 2

Further practice · Attempt part c only.

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(b) Which two skin structures are damaged by a second degree burn?

(1)

- ☒ **A** blood vessels and sebaceous glands
- ☒ **B** fat tissue and blood vessels
- ☒ **C** hair follicles and sweat glands
- ☒ **D** sebaceous glands and fat tissue

(c) Explain why a person with a second or third degree burn has a high risk of infection.

(2)

(d) Blood clotting helps to prevent infection.

Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

4. Question 5 · part c · 2 marks · Working and answer

Working and answer

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5. Question 8 · part a.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.i only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

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(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

.....

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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P 6 6 4 3 3 A 0 2 0 2 4

5. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

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6. Question 3 · part a · 2 marks

January 2022 Paper 2

Further practice · Attempt part a only.

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3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

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2 .....

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(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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6. Question 3 · part a · 2 marks · Working and answer

Working and answer

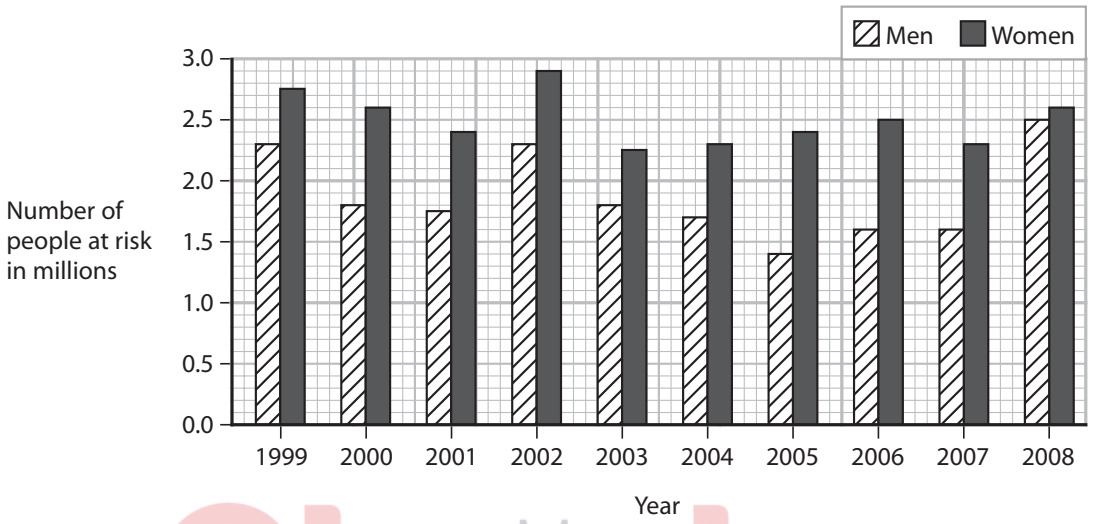
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7. Question 2 · part d · 2 marks

June 2022 Paper 2

Further practice · Attempt part d only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

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(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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P 7 0 9 7 4 A 0 4 2 0

7. Question 2 · part d · 2 marks

June 2022 Paper 2

Further practice · Attempt part d only.

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(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

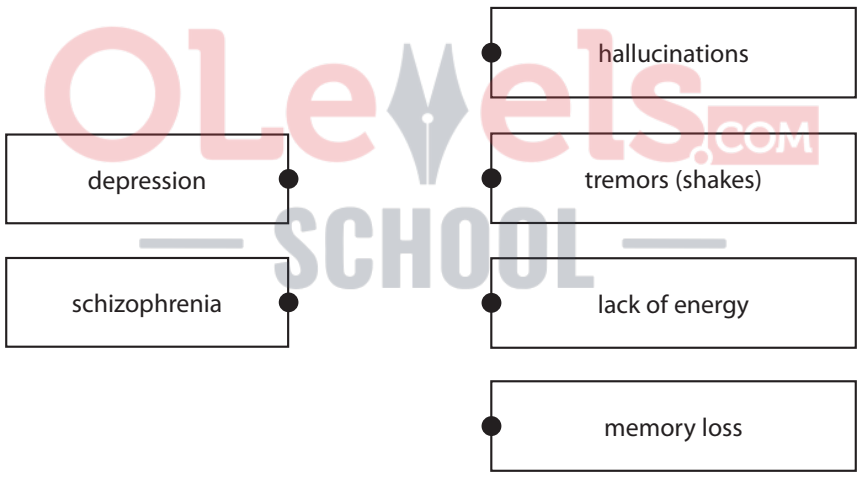
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(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

7. Question 2 · part d · 2 marks · Working and answer

Working and answer

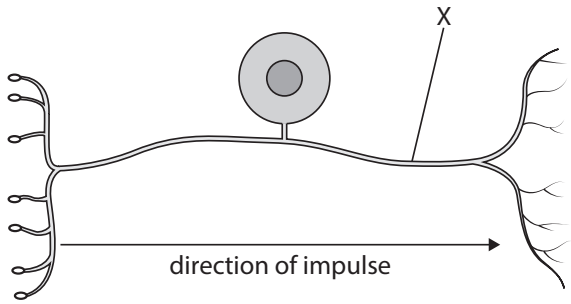
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8. Question 1 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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P 7 4 6 2 6 R A 0 2 2 4

8. Question 1 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

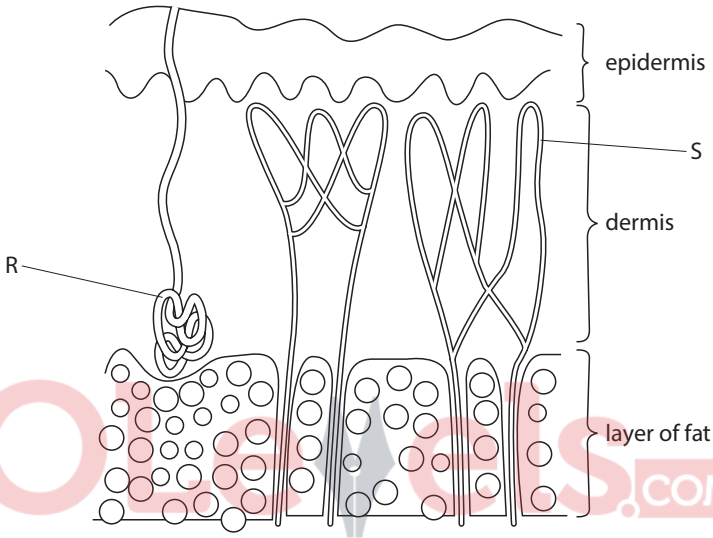
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(b) Structures in the skin respond to information from the nervous system.

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(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R .....

part S .....

(ii) Describe how part R helps to cool the body.

(2)

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8. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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